CHARACTERISTIC EQUATION

The Cayley–Hamilton Theorem

Every square matrix satisfies its own characteristic polynomial.

THEOREM

If $p(\lambda) = \det(A - \lambda I)$, then $p(A) = O$, with λ replaced by A and constant terms multiplied by I.

1. **CHARACTERISTIC POLYNOMIAL**
   $p(\lambda) = \lambda^2 - 5\lambda + 6$.
2. **SUBSTITUTE A**
   $A^2 - 5A + 6I = O$.
3. **INVERSE**
   Rearranging: $A^{-1} = \frac{1}{6}(5I - A)$, provided $\det(A) = 6 \neq 0$.
4. **HIGHER POWERS**
   Any A^k with $k \geq n$ reduces to a polynomial in A of degree at most $n - 1$.